

NATIVE BUILDER HACKATHON 2026



# AMR GUARDIAN AI

*AI-Powered Antibiotic Resistance Prediction & Clinical Decision Support*

**Team AMR Innovator's** • *Government College Woman University, Sialkot*

PRESENTED BY

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# The Growing Threat of Antimicrobial Resistance

Bacteria are evolving faster than treatment decisions are made



Lab culture results take 24–72 hours — delaying effective treatment



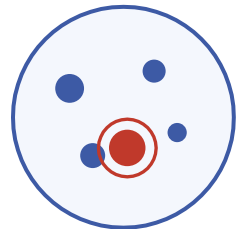
Ineffective antibiotics are often prescribed in the meantime



Misuse accelerates resistance and drives up healthcare costs



Resistant infections are rising worldwide, threatening global health



*Antibiotic-resistant colonies (red) look identical to susceptible ones under the microscope — this is the exact gap AMR Guardian AI closes with data.*



## 24–72 HRS

Typical turnaround for a lab culture & susceptibility report



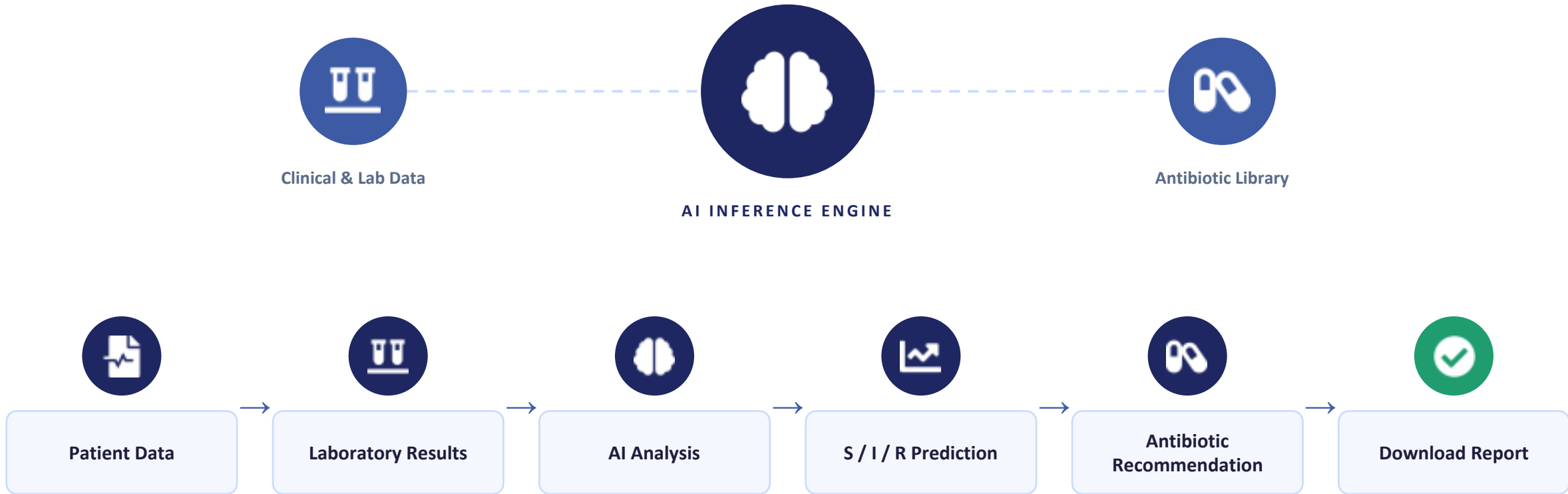
## 4.7M DEATHS

Associated with bacterial AMR worldwide in 2021 (WHO)

*Source: World Health Organization, AMR Fact Sheet, 2025*

# Our Solution: AMR Guardian AI

An intelligent platform that predicts antibiotic susceptibility in minutes, not days



*Key idea: Patient Data → AI Analysis → S/I/R Prediction → Recommended Antibiotics → Report*

# Key Features

Everything a clinician needs for a fast, confident decision



## Patient & Infection Data Input

Simple structured entry of clinical & lab data



## AI Susceptibility Prediction

Classifies each antibiotic as S, I, or R



## Confidence Scores

Transparent, quantified prediction reliability



## Antibiotic Recommendation

Suggests the most effective treatment options



## Downloadable Reports

Clinical report ready for records & handover

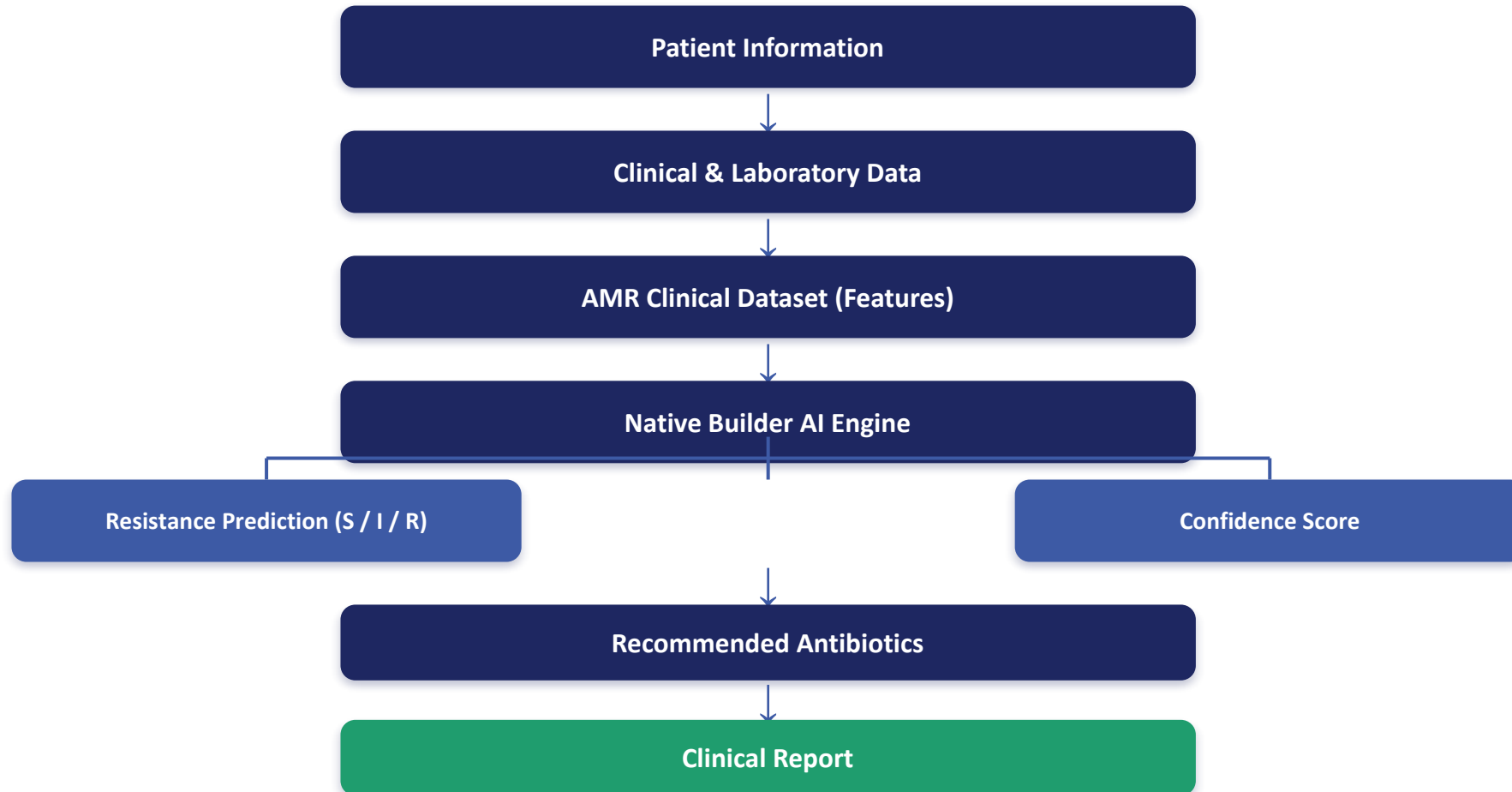


## Easy-to-Use Dashboard

Fast, intuitive web interface for any hospital

# How It Works

From patient intake to clinical report in one pipeline



# Technology Stack

Built for speed, reliability, and clinical-grade data handling



## Frontend

React + TypeScript



## Backend

Supabase



## Database

PostgreSQL (Supabase)



## AI Engine

OpenAI LLM



## Authentication

Supabase Auth



## Deployment

Native.builder



## Charts & Analytics

Recharts



# Live Demonstration

bczk3yp65b3g3god5ax1xe340.nativelyai.app



1. Enter infection & laboratory details



2. AI processes the clinical data



3. Antibiotic resistance (S/I/R) prediction



4. Recommended antibiotics displayed



5. Download the clinical report

**AMR Guardian AI**

Species: E. coli • UTI

|               |   |
|---------------|---|
| Amoxicillin   | R |
| Ciprofloxacin | S |
| Gentamicin    | I |

Confidence: 92%

Download Report

Sample prediction output

# Innovation & Uniqueness

What sets AMR Guardian AI apart from traditional methods



## Traditional Method



Manual interpretation



Slow laboratory reports



Trial-and-error treatment



Static reports



Delayed clinical decisions

## AMR Guardian AI



AI-powered prediction



Instant analysis



Data-driven recommendation



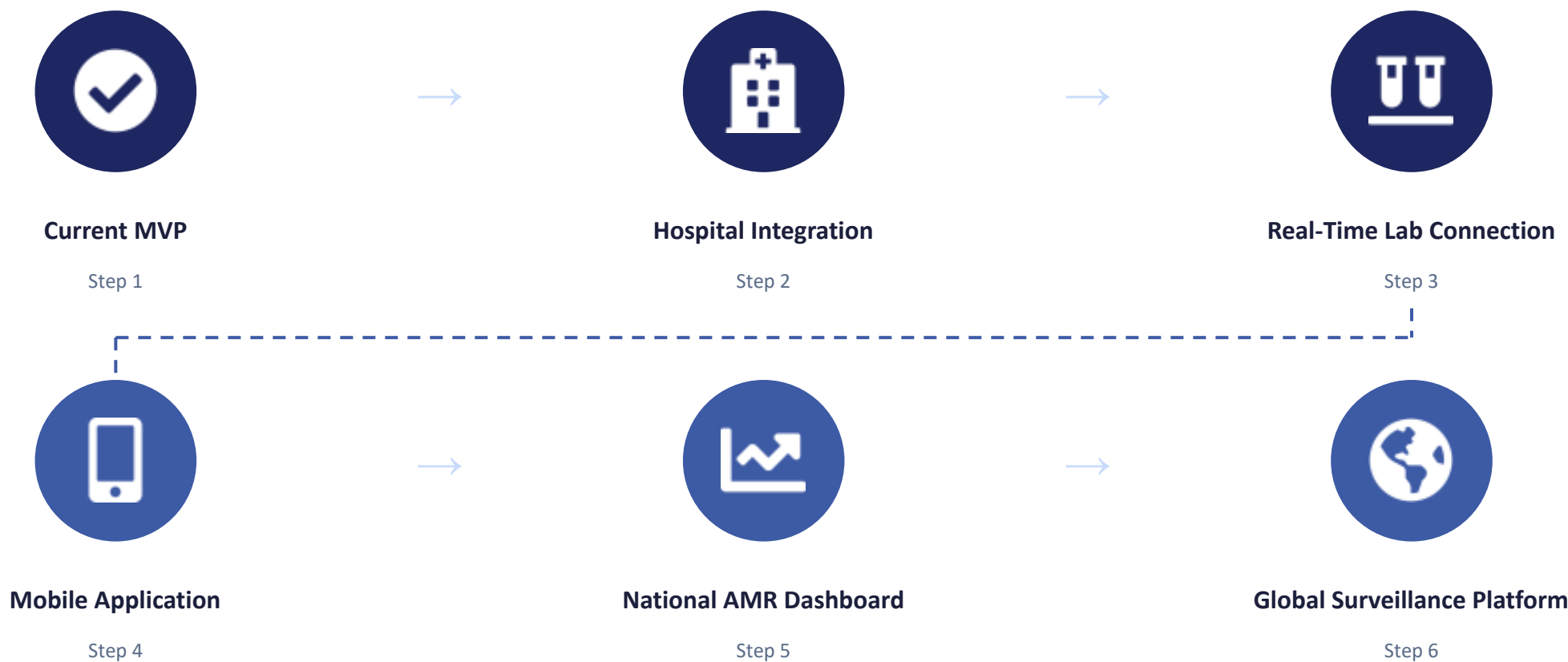
Interactive AI support



Faster clinical decisions

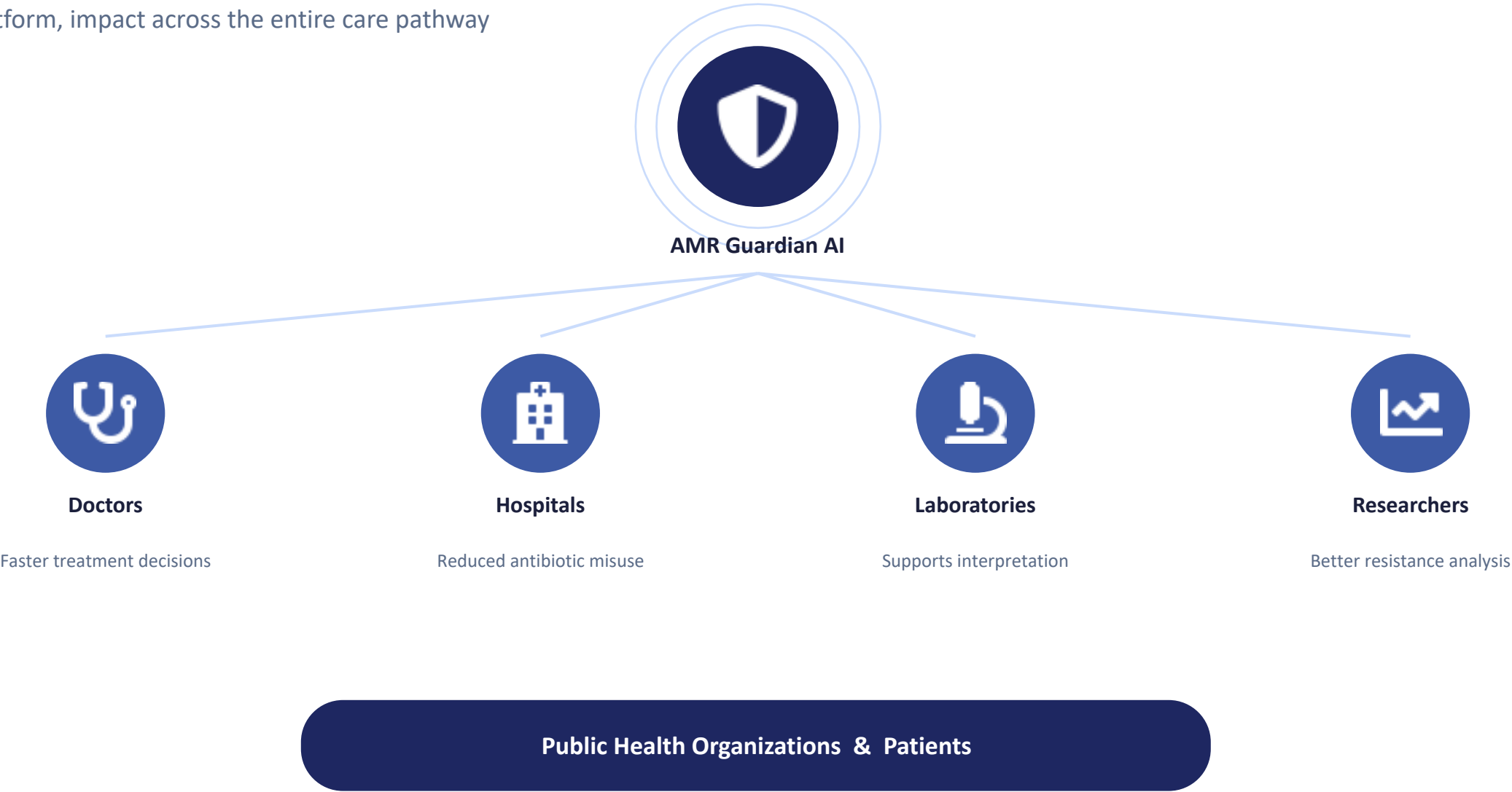
# Future Roadmap

Scaling from MVP to a global surveillance platform



# Who This Helps

One platform, impact across the entire care pathway



# Challenges & Solutions

How we addressed the hardest parts of the build



| Challenge                                                                                                           | Our Solution                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
|  Collecting reliable AMR data      |  Used curated, validated antimicrobial resistance datasets |
|  Data preprocessing                |  Cleaned and standardized clinical & lab variables         |
|  Ensuring prediction accuracy      |  AI-assisted model with iterative refinement               |
|  Clinical interpretation          |  Confidence scores support informed decisions             |
|  Building an intuitive interface |  Simple, user-friendly dashboard design                  |

# Team AMR Innovator's

Six departments, one shared mission



**Sawera Ansar**

*Biochemistry*

Team Leader, PM & UI/UX — coordinated the project, built the frontend and led documentation & submission.



**Simran Sodaghar**

*Biochemistry*

Proposed the concept, organized the clinical dataset, coordinated the AI pipeline integration, and recorded the project demo video.



**Minahil Fatima**

*Criminology*

Designed the dataset architecture — defined the 10-column MVP feature set and the S/I/R target variable.



**Sadia Eman**

*Physics*

Compiled laboratory test data (WBC, CRP, procalcitonin, cultures) for infection risk analysis and edited the project demo video.



**Laiba**

*Mathematics*

Documented the core antibiotic resistance dataset that trains the AI's S / I / R predictions.



**Bismillah Basharat**

*Biochemistry*

Data strategy lead — selected 9 key clinical features and evaluated open-access medical datasets.



# Bringing It All Together

AMR Guardian AI combines artificial intelligence with antimicrobial resistance data to help clinicians choose the right antibiotic, faster.



**Faster antibiotic selection**



**Reduced antimicrobial resistance**



**Improved patient outcomes**



**Smarter antibiotic stewardship**

## Our Vision

*To make AI-assisted antimicrobial resistance prediction accessible to hospitals, laboratories, and healthcare professionals worldwide.*





# Thank You!

*"Fighting Antibiotic Resistance with Artificial Intelligence."*

**Questions?**

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